ENGINEERING SYSTEMS / HUMAN-IN-THE-LOOP ROBOTICS

AETHER-6

Astro Robot: a private AI-connected hexapod for GNC experimentation and aerospace learning

Technical project paper | Emanuele Magno | 23 October 2026

Abstract

Astro Robot is a Raspberry Pi-based six-legged robotics prototype that connects a Freenove FNK0052 hexapod to a private portfolio environment. The system combines supervised low-speed teleoperation, live robot telemetry, an experimental attitude-control stack, and two owner-only AI workspaces: Astro Tutor for study support and Astro Codex for repository-aware engineering assistance. The language model is kept outside the actuator path: physical motion requires an explicit owner command, while the Pi enforces readiness, bounded commands, stop behavior, and configured safety checks. The project is an educational engineering prototype, not an autonomous or flight-qualified vehicle.

Project thesis: one private interface links a physical robot, an inspectable GNC experiment, and aerospace learning tools - with the human operator remaining in command.

At-a-glance

ELEMENT	PROJECT DEFINITION
Platform	Freenove Big Hexapod Robot Kit FNK0052, Raspberry Pi 4, MPU6050 IMU, vendor servo-control board and an Astro software layer.
Control path	Astro Pi API over a private Tailscale link; current portfolio integration uses the Freenove command server locally on the Pi for locomotion and body-attitude requests.
AI modules	Astro Tutor and Astro Codex are private, owner-only portfolio tabs. They advise and analyze; they do not issue autonomous robot motion commands.
Validation status	Manual walking/turning, servo motion, IMU telemetry and GNC commands have been observed during owner-run tests. Closed-loop stabilization still exhibits tremor and occasional saturation; it is not claimed as validated.

Keywords: hexapod, GNC, MEKF, LQR, teleoperation, robotics education, human-in-the-loop AI.

1. System architecture and capabilities

Astro is split across three trust zones. The robot and its API run on the Raspberry Pi; the portfolio backend relays a narrow, authenticated set of owner actions through the private tailnet; the browser renders telemetry and the human-facing tools. This separation keeps the AI conversation path distinct from the motion-control path.

ROBOT / FNK0052	ASTRO PI / CONTROL	PRIVATE PORTFOLIO
18 leg servos PCA9685 boards MPU6050 IMU Optional sonar / camera	Astro API + Freenove TCP adapter Attitude estimator and LQR Readiness checks, dead-man, STOP	Owner-only robot panel Astro Tutor Astro Codex Live telemetry / 3D view

Integrated feature set

CAPABILITY	BEHAVIOR / STATUS
Supervised locomotion	Hold-to-move controls for forward/reverse, lateral movement and yaw; release/STOP ends the command. Commands are bounded by the portfolio adapter and Pi-side dead-man behavior.
GNC commissioning	Monitor and stabilize modes, stationary gyro calibration, roll/pitch references, status/fault telemetry and a bounded controller output. Stabilization remains experimental pending repeatable physical validation.
Robot telemetry	Controller state, actuation arm state, IMU attitude/rates, GNC commands and safety/readiness status. Sonar and battery fields depend on hardware and driver availability.
3D visualization	Browser view driven by reported attitude and commanded gait. The included PhantomX reference model is not the exact FNK0052 geometry and does not show measured individual servo-joint angles.
Optional hardware	Camera support is optional; camera-dependent face tracking/orbit behaviors are disabled unless the camera and locked pan/tilt hardware are commissioned. Sonar can be optional, while a fresh near-obstacle reading can still block forward movement.
Voice and assistant	The Pi stack documents wake-word, speech-to-text, text/voice chat and speech output. Local inference is supported by the software design; an optional cloud LLM route is configuration-dependent and is not part of the actuator loop.
Mission tools	Bounded teleoperation recording/replay and QR waypoint-label reading are software features; QR reading identifies a marker and is not SLAM or autonomous navigation.

Robot commands are direct, owner-triggered actions. Astro Tutor/Codex responses do not themselves control locomotion, servo outputs, or GNC mode.

2. Custom GNC stack

Astro adds a custom roll/pitch stabilization chain around the Freenove hexapod. In the current integrated path, Astro reads the MPU6050 directly for its own state estimate, while high-level body-attitude requests are sent through the Freenove command interface. Freenove vendor balance is a separate controller and must not run concurrently with Astro LQR.

Signal and control flow

STAGE	IMPLEMENTATION
Sensing	MPU6050 accelerometer + gyroscope; configurable sensor-axis permutation, sign mapping, and level-frame correction.
Navigation	Multiplicative / error-state attitude EKF: quaternion attitude, gyro-bias state, gyro propagation, accelerometer gravity update for roll/pitch; yaw is gyro-propagated by default.
Guidance	Bounded, rate-limited roll/pitch references. A reference can target level or hold a captured attitude within configured limits.
Control	Discrete LQR with state x = [roll error, pitch error, roll rate, pitch rate] and control u = -Kx. The gain is computed at startup from a discrete Riccati solution and a configurable local plant model.
Actuation	Bounded roll/pitch body-attitude correction sent as a high-level kinematic request to the Freenove motion/IK path. This is not per-foot torque control or a whole-body force allocator.
Supervision	Actuation is gated by calibration, fresh sensor data, safety envelopes and explicit mode selection. Faults zero the correction and return the controller to Monitor; output is limited by a configured command bound.

Reported commissioning configuration

The owner-provided telemetry showed an approximately 25 Hz GNC loop, a 4 degree maximum command, and LQR parameters ASTRO_GNC_LQR_ACTUATOR_GAIN=10.0 and ASTRO_GNC_LQR_DAMPING=2.4. These are prototype configuration values, not independently optimized gains.

Commissioning interpretation

The MPU6050 must be calibrated while the robot is stationary, and its axes/signs and level offsets must match the robot body frame. A controller-active flag confirms software execution, not successful mechanical stabilization. Direction must be verified on a supported, unloaded bench before floor operation. Gait-aware stabilization and slope walking require separate validation.

Observed limitation: owner-run tests show the robot can tremble on a level surface in Stabilize mode, and LQR saturation has occurred during some tests. Therefore closed-loop physical stability is experimental and is not presented as solved.

3. Astro Tutor, Astro Codex, and the owner interface

The portfolio adds a private human interface around the physical system. Astro Tutor and Astro Codex remain accessible only to the site owner. The public Neural Core chat remains separate. A robot panel is rendered inside the private owner tools, and only the owner can query telemetry or issue the allowlisted controls.

Astro Tutor - aerospace learning companion

Tutor supports explanations, quizzes, flashcards and study plans grounded in the owner's Knowledge Nodes and indexed course material. When the owner chooses "Add telemetry to Tutor question," selected robot telemetry is placed into a draft prompt for review; the owner sends it. Tutor can then help interpret estimates, faults and control behavior without obtaining actuator authority.

Astro Codex - repository engineering copilot

Codex can inspect selected GitHub repositories through a narrowly scoped text-file context, review code and discuss implementation. Repository content is treated as untrusted input. A proposed change can become a validated patch and a draft pull request only after explicit owner confirmation; the default branch is not written directly. The Codex integration does not execute arbitrary repository code on the Pi.

Owner robot panel

UI ELEMENT	FUNCTION
Link and readiness	Check private Pi connection and display whether the controller, IMU, GNC and motion prerequisites are available.
Manual control	Hold-to-move controls with release-to-stop behavior, explicit STOP, bounded motion/yaw requests, and Pi-side readiness/dead-man checks.
GNC commissioning	Choose Monitor/Stabilize, calibrate stationary gyro, set roll/pitch references, and inspect active/fault/saturation telemetry.
Other explicit actions	Separate Freenove balance control requires confirmation; RGB LED presets and supported body-pose actions are allowlisted.
Live view	Poll telemetry and drive a browser 3D reference model; telemetry can be shared into Tutor only after owner review.

Security and control boundary

The browser never receives the Pi bearer token. Django accepts only private Tailscale origins, removes redirects, limits telemetry fields and exposes fixed actions rather than arbitrary Pi paths. Motion requests are bounded; stop remains available even when readiness blocks motion. The LLM is not a robot command parser and cannot autonomously move Astro.

Human-in-the-loop contract: the owner reviews Tutor context, chooses Codex repository actions, selects GNC mode, and directly initiates robot movement.

4. Results, limitations, and licensing

Results are summarized from owner-reported commissioning logs and operation. They are integration evidence, not a controlled experiment or independent verification.

AREA	OBSERVED RESULT	EVIDENCE / LIMIT
Robot link	Private Pi API and portfolio bridge have connected over Tailscale.	Successful health/telemetry responses reported by owner; no public robot endpoint.
Servos / gait	Servo motion, forward walking and yaw rotation have been observed.	Qualitative owner-run tests. Gait speed/trajectory and chassis clearance still need tuning.
IMU / GNC	MPU6050 telemetry, gyro calibration and bounded attitude commands have been observed; one commissioning log reported a loop rate near 25 Hz.	Tremor on a level surface and occasional LQR saturation are unresolved; active status alone does not certify stability.
Portfolio AI	Owner-only Tutor/Codex and robot telemetry panel are integrated in the portfolio code.	Tutor/Codex assist the owner; they do not autonomously command the robot.
3D / camera	A live-attitude 3D reference view is implemented; camera path is optional.	Reference model is not the FNK0052 CAD model; individual joint telemetry and camera operation are not established in the present commissioning record.

Limitations and next validation gates

The robot is a small ground prototype with high-level servo kinematics. The present evidence does not establish repeatable disturbance rejection, level-surface stability, slope traversal, autonomous navigation, obstacle avoidance beyond the configured forward stop, battery endurance, or safety certification. Next tests should record synchronized roll/pitch, rates, LQR commands, saturation, actuator state and gait; verify sign and neutral pose with legs unloaded; then evaluate supported static correction before any short, supervised floor test.

Freenove hardware and license notice

The physical chassis is the Freenove Big Hexapod Robot Kit FNK0052, a third-party product; this paper does not claim ownership of or relicense the hardware design, Freenove name or logo. Freenove states that files in its FNK0052 Raspberry Pi repository are distributed under Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported (CC BY-NC-SA 3.0). Astro’s third-party notice identifies Freenove-derived hardware mapping details, including PCA9685 addresses, leg channels and linkage dimensions. Any adapted Freenove material must retain attribution, non-commercial use and share-alike obligations. Astro’s original software is separately licensed under MIT; third-party terms remain with their respective assets.

Freenove source: official FNK0052 repository and LICENSE.txt. License deed: CC BY-NC-SA 3.0.

Project references

Astro Robot Stack v0.8.0 work: README, ARCHITECTURE, GNC_SELF_STABILIZATION, FEATURES_V090, FREENOVE_INTEGRATION and THIRD_PARTY_NOTICES. AstroX portfolio bridge: source/deploy/astro-robot-bridge/README.md, astro_robot.py and astro_robot_views.py. Freenove module-test and assembly guidance: Module Test and Hexapod Robot.

PORTFOLIO PROJECT CARD

AETHER-6

A private AI-connected hexapod for aerospace learning and GNC experimentation

PROJECT NAME

AETHER-6 - Astro Robot

DESCRIPTION

A Raspberry Pi 4-controlled Freenove FNK0052 hexapod connected to a private web portfolio. The platform combines supervised locomotion, live telemetry, an experimental custom GNC controller, Astro Tutor study tools, and Astro Codex repository assistance. The operator stays in command of every physical movement.

CHALLENGE

Build a useful bridge between a real six-legged robot, aerospace engineering study and software development while keeping the robot reachable only over a private network. The system also needed to expose sensor and controller state clearly, so that GNC experiments could be inspected rather than hidden behind an “AI autopilot” claim.

APPROACH

Use a Raspberry Pi API as the robot-side boundary; connect the Freenove motion server locally; read MPU6050 telemetry for Astro’s own attitude estimator; process attitude through a multiplicative/error-state EKF, rate-limited roll/pitch guidance and discrete LQR; send bounded body-attitude requests through Freenove kinematics. A private Django bridge exposes only owner-authorized telemetry and fixed, bounded actions. Tutor adds reviewed telemetry to learning questions; Codex inspects authorized GitHub text and requires a separate confirmation for a draft PR.

RESULTS

Owner-run tests have demonstrated servo movement, manual forward gait and yaw rotation, private Pi connectivity, IMU telemetry, gyro calibration, GNC mode changes and attitude-command output. Reported GNC telemetry reached approximately 25 Hz. Physical stabilization is still experimental: tremor on a level surface and occasional LQR saturation remain unresolved. The 3D view uses a reference hexapod model, not an exact FNK0052 digital twin; the project makes no claim of autonomous slope walking or flight readiness.

Portfolio maturity label: Integrated prototype | Demonstrated manual motion and telemetry | GNC stabilization under active validation

Suggested one-line summary

“AETHER-6 turns a hobby hexapod into a private, inspectable GNC and aerospace-learning testbed - with a human operator, not an AI model, in control.”

Prepared in English from project files and owner-reported test logs available on 23 October 2026. Performance claims are intentionally limited to the evidence described above.

Engineering theory for implementation

Appendix A | Frames, sensing, attitude estimation and control model

This appendix adds the mathematical foundation behind the prototype. It distinguishes algorithms present in the Astro GNC v0.9 source from the currently integrated portfolio deployment, which has been observed using the Freenove TCP servo server plus direct MPU6050 reads. A formula describes a model; it does not by itself establish that the model matches the physical robot.

A.1 Frames and inertial sensing

Define a right-handed body frame B fixed to the chassis and a world frame W with +z upward. The MPU6050 frame S can be rotated relative to B; first apply a signed axis permutation, then a fixed mounting rotation. Accelerometer readings include gravity and dynamic acceleration; gyro readings contain angular rate and bias.

```
omega_B = R_BS (omega_S - b_g) ; f_B = R_BS f_S
```

Here R_BS is the calibrated sensor-to-body rotation and b_g is gyro bias. A stationary calibration estimates b_g by averaging samples. It does not determine the chassis level by itself; that requires a verified level reference and a separate mounting-offset calibration.

A.2 Quaternion attitude model

The estimator stores a unit quaternion q_BW for body-to-world orientation and a three-axis gyro bias. With body angular rate omega, continuous quaternion kinematics are:

```
q_dot = 1/2 q tensor [0, omega_B] ; ||q|| = 1
```

A discrete implementation integrates the rotation vector delta_theta = omega_B dt, composes it with q, and renormalizes. The error-state is delta_x = [delta_theta, delta_b_g]. Locally, the covariance prediction uses a linearized error model Phi approximately I + F dt and process noise Q_d.

```
P(k+1|k) = Phi P(k|k) Phi^T + Q_d
```

The normalized accelerometer vector is treated as a gravity-direction observation only when its magnitude lies inside a configured gate around 1 g. Predicted gravity in body coordinates is h(q) = R_BW^T [0,0,1]^T; the residual is r = z_acc - h(q). The Kalman update computes K = P H^T (H P H^T + R)^-1, applies a small-angle correction to q and a bias correction, then uses the Joseph covariance form. During strong linear acceleration, gravity is not a clean attitude observation, so gating and covariance inflation matter.

```
delta_x = K r ; P+ = (I-KH)P(I-KH)^T + K R K^T
```

Observability note: gravity constrains roll and pitch in quasi-static conditions, but not yaw. With no trusted magnetometer or external heading source, yaw drifts under gyro integration. This is expected and must not be reported as absolute heading.

Implementation status: the Astro MEKF source implements quaternion propagation, gyro-bias estimation, accelerometer norm gating and numerical gravity Jacobians. The Freenove TCP integration does not provide raw IMU samples; Astro reads the MPU6050 separately.

Control theory and GNC loop

Appendix B | Guidance, discrete LQR, saturation and supervisory logic

B.1 Small-angle plant around a supported operating point

For low roll and pitch near a fixed stance, approximate each axis as a damped rotational mode. The Astro LQR implementation models the effect of a high-level body-attitude request as an equivalent angular acceleration. This is an identified/configurable surrogate, not a torque-level model: the Freenove command interface hides joint torques, ground reactions and servo dynamics.

```
x = [phi-phi_ref, theta-theta_ref, p, q]^T
x_dot = A_c x + B_c u
A_c = [[0,0,1,0],[0,0,0,1],[0,0,-d,0],[0,0,0,-d]]
B_c = [[0,0],[0,0],[g,0],[0,g]] ; u = [u_roll,u_pitch]^T
```

The source discretizes with forward Euler at sample time dt: A = I + A_c dt and B = B_c dt. At 25 Hz, dt is approximately 0.04 s. A more exact zero-order-hold discretization would use matrix exponentials; the implementation does not do that.

B.2 LQR objective and Riccati equation

The controller chooses a bounded command to reduce attitude/rate error while penalizing excessive actuator requests. For the discrete linear model, the infinite-horizon quadratic objective is:

```
J = sum(k=0..infinity) [x_k^T Q x_k + u_k^T R u_k]
```

The discrete algebraic Riccati equation (DARE) and optimal feedback are:

```
P = A^T P A - A^T P B (R+B^T P B)^-1 B^T P A + Q
K = (R + B^T P B)^-1 B^T P A ; u* = -K x
```

The implementation solves the DARE iteratively with NumPy, then clips each command to the configured limit. Clipping is not anti-windup; this controller has no integral state, so it avoids integral windup but can still saturate and lose correction authority. The reported setup used about 25 Hz, a 4 degree command bound and configurable actuator-gain/damping values (10.0 and 2.4 in one owner-provided configuration). Those values are starting model parameters, not validated optima.

```
u = clip(-Kx, -u_max, +u_max)
```

B.3 Guidance and supervisory safety

Guidance rate-limits references so a sudden UI change does not become an instantaneous body-pose step. The source defaults include a small reference envelope (about +/-8 degrees) and 4 degrees/second slew. A separate supervisor blocks actuation when the mode is not Stabilize, calibration is missing, sensor data are stale, state values are invalid, the attitude/rate leaves its envelope, or a disallowed gait is active. A fault returns control to Monitor and zeros the correction.

```
|phi|, |theta| <= phi_max ; |p|, |q| <= omega_max
```

Stability is a property of the full closed loop, including estimator delay, communication timing, servo response, linkage, contacts and battery supply. The LQR flag "controller active" means the software supervisor permits the loop; it is not evidence that the physical plant is stable. Tremor on a level surface and occasional saturation have been observed, so the hardware loop remains experimental.

Recommended identification: record sensor, reference, command and saturation; perturb the supported chassis by a small known angle; estimate the local response and delay; fit the plant; tune Q/R conservatively; then repeat tests with gait disabled before any walking trial.

Hexapod mechanics and locomotion

Appendix C | Coordinate transforms, inverse kinematics, gait and support stability

C.1 Leg geometry and inverse kinematics

A six-leg platform has six leg-base frames. For a foot target p_i in body coordinates, transform the point into leg-local coordinates using the inverse body rotation and the leg mounting angle. A three-joint leg typically has coxa yaw alpha, femur pitch beta and tibia pitch gamma. The source direct driver documents nominal link lengths of 33, 90 and 110 mm.

```
p_i^L = R_LB ( R_BW p_i^W - t_LB )
```

After removing the coxa offset, let the remaining planar distance be r and vertical offset be z. The femur-tibia triangle gives a knee angle through the cosine rule:

```
D = (r^2 + z^2 - L_f^2 - L_t^2) / (2 L_f L_t)
gamma = pi - arccos( clamp(D, -1, 1) )
beta = atan2(z,r) - atan2(L_t sin(gamma), L_f + L_t cos(gamma))
```

The coxa angle is obtained from the horizontal foot bearing and the local geometry. Mirror signs differ between left/right or forward/rearward legs. Reachability requires |D| <= 1 and every joint angle must remain inside measured mechanical limits. Saturating an unreachable request silently distorts the foot path; a robust controller should detect and reject it before PWM output.

C.2 Tripod gait and support polygon

A common hexapod gait alternates two tripods, for example legs {1,3,5} and {2,4,6}. The duty factor beta is the fraction of a cycle that a foot remains on the ground; beta > 0.5 provides overlapping support in a quasi-static tripod gait. During stance, the foot moves backward relative to the body to create forward body motion; during swing, it lifts and advances.

```
phi_i = (omega_g t + phase_i) mod 2pi
z_i(phi) = 0 during stance; z_i(phi) = h sin(pi s) during swing, 0<=s<=1
```

For quasi-static stability, project the center of mass onto the support polygon formed by stance feet. A simple stability margin is the minimum signed distance from that projection to any polygon edge:

```
m_s = min_j distance( projection(COM), edge_j )
```

Positive margin is a useful geometric condition for a static stance, but it does not guarantee dynamic stability, traction or servo torque reserve. Body height and foot clearance must be checked across the entire gait cycle, not just at the endpoints.

C.3 What the software actually does

The source direct-I2C Astro driver contains body transforms, 3-link inverse kinematics and a bounded alternating-phase gait. The portfolio deployment discussed in the main paper currently routes high-level locomotion through Freenove TCP; its actual gait generation is vendor-side. The appendix equations explain the design needed for independent control, but should not be confused with a verified deployed tripod implementation. Chassis scraping, gait asymmetry and slow walking have been reported in owner tests and remain mechanical/control tuning issues.

To commission safely: verify leg order and mirrored signs; lift/support the chassis; command small joint changes; confirm all eighteen servo channels; measure neutral stance and foot clearance; test one gait phase at a time; inspect current draw and temperature; then test on a level floor at low speed.

Actuation, embedded systems and verification

Appendix D | PWM, power, timing, network boundaries, test design and AI integration

D.1 PWM servo command and power budget

The PCA9685 provides 12-bit phase/count PWM. At frequency f_PWM, one count spans T_PWM/4096. For a pulse width t_pulse, the approximate off-count is:

```
N_off = round(4096 * t_pulse / T_PWM) = round(4096 * t_pulse * f_PWM)
```

For example, at 50 Hz the period is 20 ms; a nominal 1.5 ms center pulse corresponds to about 307 counts. The source direct driver maps a nominal 0.5-2.5 ms pulse across 0-180 degrees. Real servos vary: establish safe endpoints experimentally, then limit every joint to a conservative calibrated range. Never assume the mechanical linkage can use the full electrical angle.

Servo stalls produce large current transients. The battery, board regulator, wiring and ground return must support simultaneous servo peaks without brownout or I2C resets. The battery-voltage ADC is only a proxy for available torque; it does not report per-servo current or thermal state. Keep logic and servo power correctly distributed and retain a physical LOAD cutoff.

D.2 Timing, communications and safety boundary

The Pi runs a local robot API; the portfolio backend reaches it over the private Tailscale link. The browser never receives the bearer token. The deployed adapter forwards a narrow allowlist to the Pi, while the Pi bounds motion, requires explicit actuation arming and provides STOP/dead-man behavior. The Freenove command TCP port should remain local to the Pi, not exposed publicly. Network transport is supervisory; the time-critical estimator/controller should run locally on the Pi.

```
end-to-end delay = sensing + estimation + control + transport + actuator response
```

At 25 Hz, the nominal control period is 40 ms. A stale-sensor threshold, bounded request rate and watchdog are necessary because Wi-Fi/Tailscale latency is variable. If a command stream stops, the Pi should transition to a safe stop; "HTTP 200" confirms receipt/handling, not successful leg motion. Optional sonar/camera are separate from basic actuation, but any obstacle stop logic must use fresh, valid readings.

D.3 Validation that turns a prototype into evidence

METRIC	DEFINITION / USE
Attitude RMS	sqrt(mean((angle - reference)^2)); evaluate roll and pitch separately over a stated interval.
Settling time	Time after a defined disturbance until attitude enters and stays within a stated error band.
Peak error	max |angle-reference|; report disturbance amplitude and initial stance.
Saturation fraction	N_sat / N_samples; persistent clipping indicates insufficient authority or a poor model.
Gait clearance	Minimum measured chassis-to-ground distance over repeated cycles, with payload and surface specified.
Reliability	Successful cycles / attempted cycles, plus faults, battery state and test duration.

Use staged tests: software simulation/unit tests; stationary sensor calibration; supported actuator direction test; supported closed-loop test; unloaded gait test; low-speed floor test; repeat under documented battery and surface conditions. Change one parameter at a time and preserve a black-box log. Report sample count, loop rate, sensor frame, reference, command limits and failure criteria. Do not infer a safety margin from a single successful run.

D.4 Astro Tutor and Astro Codex in the engineering loop

Astro Tutor can explain course concepts and interpret owner-selected telemetry or indexed notes; Astro Codex can inspect the selected Astro Robot repository and help review code. Both are private portfolio tools and remain outside the control loop. They can help formulate hypotheses, derive equations and summarize tests, but the operator must verify code, choose GNC mode and initiate every physical action. AI-generated advice is not a safety interlock, controller, or source of validated gains.

Hardware and license provenance: the robot's physical frame/structure, leg-linkage design and kit electronics are Freenove FNK0052 hardware; Astro contributes the custom software, GNC experiments, private portfolio integration and learning tools, and does not claim Freenove's mechanical design as original Astro work. The Freenove software repository is published under CC BY-NC-SA 3.0 Unported. This applies to attributed upstream code/adapted hardware mapping under its terms, not ownership of the physical kit or unrelated Astro code. Astro's own license remains as stated in its repository. Review notices before redistribution or commercial use.

Sources used for these appendices: Astro Robot Stack v0.8.0 worktree, docs/GNC_SELF_STABILIZATION.md, docs/FREENOVE_INTEGRATION.md, docs/ARCHITECTURE.md, THIRD_PARTY_NOTICES.md, and pi/astro_robot/gnc plus robot drivers. Equations are explanatory summaries of these sources and standard rigid-body/control theory; the distinction between source reference and currently integrated deployment is intentional.